

Notes: Lou (RFS, 2012)
A Flow-Based Explanation for Return Predictability

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1 Big picture

- **Question.** Can predictable retail flows into and out of mutual funds generate return predictability in both stocks and mutual funds?
- **Core idea.** Mutual fund investors chase past performance. These predictable flows force funds to buy or sell their existing holdings. If arbitrage capital is limited, the resulting demand pressure moves prices temporarily.
- **Unifying mechanism.** The same flow-induced trading mechanism can help explain three empirical facts: short-run mutual fund performance persistence, the “smart money” effect, and stock price momentum.
- **Main empirical object.** The paper constructs stock-level *flow-induced trading*, FIT , by aggregating the predicted trades of all mutual funds that hold a given stock.
- **Expected flow pressure.** Because fund flows are predictable from past fund performance and lagged flows, the paper constructs expected flow-induced trading, $E[FIT]$, and a fund-level version, $E[FIT^*]$.
- **Main result.** Expected flow-induced trading positively predicts stock and fund returns over the next year, but the effect reverses in later years. This pattern is exactly what a temporary price pressure story predicts.
- **Economic takeaway.** Some predictability that looks like skill, investor intelligence, or underreaction may instead reflect predictable nonfundamental demand shocks created by delegated investment flows.

2 Incremental contribution of the paper

- **From realized fire-sale pressure to expected price pressure.** Earlier work shows that realized mutual fund flows can move stock prices. This paper asks whether the predictable part of such flow pressure forecasts future returns.
- **A stock-level aggregate pressure measure.** The paper builds FIT by aggregating flow-induced trades across all funds holding each stock, rather than examining only extreme fund flows or aggregate market flows.
- **A fund-level translation.** The paper maps stock-level $E[FIT]$ into a fund-level measure, $E[FIT^*]$, by taking the portfolio-weighted average of expected pressure across each fund’s holdings.
- **A unifying explanation for mutual fund anomalies.** Performance persistence and the smart money effect are shown to be largely subsumed by $E[FIT^*]$ in portfolio tests and Fama-MacBeth regressions.
- **A partial explanation for stock momentum.** Flow-induced trading explains a meaningful share of stock price momentum, especially after 1993 and among large-cap stocks, where mutual funds are more important.
- **A comovement implication.** The paper goes beyond first moments: stocks exposed to similar expected mutual fund flow pressure comove more in subsequent returns.

3 Data and measurement

3.1 Mutual fund data

- Quarterly mutual fund holdings come from CDA/Spectrum from 1980 to 2006.
- Fund returns, total net assets, expense ratios, and other fund characteristics come from the CRSP survivorship-bias-free mutual fund database.
- CDA/Spectrum and CRSP are merged using MFLinks.
- The sample focuses on domestic equity mutual funds and excludes international funds, fixed-income funds, and precious-metal funds.
- The final sample contains 77,983 fund-quarter observations and 2,989 distinct mutual funds.

3.2 Fund flows

Quarterly fund flow is measured as

$$flow_{i,t} = \frac{TNA_{i,t} - TNA_{i,t-1}(1 + RET_{i,t}) - MGN_{i,t}}{TNA_{i,t-1}}, \quad (1)$$

where $MGN_{i,t}$ adjusts for increases in total net assets due to fund mergers.

Interpretation. The flow measure isolates new investor money after adjusting for fund returns and mergers. Positive values indicate investor inflows; negative values indicate redemptions.

3.3 Stock data

- Stock returns and trading data come from CRSP.
- The paper excludes stocks with prices below five dollars and stocks in the bottom NYSE size decile to reduce microstructure concerns.
- Liquidity measures are taken from Hasbrouck's data; the main reported measure is the Gibbs estimate of effective bid-ask spreads from the Basic Market-Adjusted model.

4 Models and empirical specifications

4.1 Fund trading response to capital flows

The paper first estimates how fund managers trade when investors send money in or withdraw money out:

$$trade_{i,j,t} = \beta_0 + \beta_1 flow_{i,t} + \gamma_2 X + \gamma_3 flow_{i,t} X + \epsilon_{i,j,t}. \quad (2)$$

- $trade_{i,j,t}$ is the percentage change in fund i 's shares held of stock j .
- $flow_{i,t}$ is the fund's capital flow.
- X includes stock- and portfolio-level trading cost variables, especially ownership share and effective bid-ask spread.

Interpretation. If funds perfectly scale their old portfolios, β_1 should be one. Deviations from one reveal liquidity constraints, cash use, or new-position creation.

4.2 Flow-induced trading

Stock-level flow-induced trading is

$$FIT_{j,t} = \frac{\sum_i shares_{i,j,t-1} \cdot flow_{i,t} \cdot PSF_{i,t-1}}{\sum_i shares_{i,j,t-1}}, \quad (3)$$

where PSF is the partial scaling factor estimated from the trading-response regressions.

Interpretation. FIT measures how much buying or selling pressure a stock receives because the funds that hold it experience investor flows.

4.3 Expected flow-induced trading

The expected version replaces realized flows with predicted flows:

$$E_t[FIT_j] = \frac{\sum_i shares_{i,j,t} \cdot E_t[flow_i] \cdot PSF_{i,t}}{\sum_i shares_{i,j,t}}. \quad (4)$$

The fund-level version is the portfolio-weighted average of stock-level expected pressure:

$$E_t[FIT_i^*] = \sum_j E_t[FIT_j] \omega_{i,j,t}. \quad (5)$$

Interpretation. $E[FIT]$ is expected stock-level price pressure. $E[FIT^*]$ is the expected price pressure embedded in a mutual fund's current holdings.

4.4 Predicting fund flows

The flow prediction regression is

$$flow_{i,t+1} = \beta_0 + \beta_1 alpha_{i,t} + \beta_2 adjret_{i,t} + \beta_3 flow_{i,t} + \beta_4 flow_{i,t-1} + \beta_5 flow_{i,t-2} + \beta_6 flow_{i,t-3} + \epsilon_{i,t+1}. \quad (6)$$

Interpretation. Past abnormal fund performance and past flows predict future fund flows, which makes future price pressure predictable.

4.5 Horse-race regressions

For fund performance, the central Fama-MacBeth specification is

$$RET_{i,t+1} = \beta_0 + \beta_1 E_t[FIT_i^*] + \beta_2 alpha_{i,t} + \beta_3 flow_{i,t} + \Gamma' Controls_{i,t} + \epsilon_{i,t+1}. \quad (7)$$

For stock momentum, the core specification is

$$ret_{j,t+1:t+3} = \beta_0 + \beta_1 E_t[FIT_j^k] + \beta_2 ret_{j,t-k:t-1} + \Gamma' Controls_{j,t} + \epsilon_{j,t+1:t+3}. \quad (8)$$

Interpretation. These regressions ask whether conventional predictors retain power after controlling for flow-induced price pressure.

5 Identification and empirical design

5.1 What the design identifies

- The paper identifies a predictive relation between expected mutual fund flow pressure and subsequent returns.
- The reversal pattern is crucial: positive short-run returns followed by negative long-run returns are consistent with temporary price pressure, not permanent information.
- The paper does not claim that every flow is exogenous. Instead, it focuses on the predictable component of flow-induced trading and shows that its return pattern matches price-pressure predictions.

5.2 Why the mechanism is plausible

- Fund flows are strongly predictable from past performance and lagged flows.
- Fund managers substantially scale existing holdings in response to flows, especially for redemptions.
- Flow pressure aggregated across many funds produces large enough stock-level demand shocks to move prices.
- Reversals in years two and three suggest that the short-run return continuation is not purely information-based.

5.3 Main limitations

- Flow-induced trading may be correlated with omitted information if funds with similar holdings receive flows for information-related reasons.
- The decomposition between flow-induced and information-motivated trading is empirical, not directly observed.
- The results are strongest in settings where mutual funds are important holders, especially large-cap stocks and the later sample period.

6 Tables and figures

6.1 Table 1: Summary statistics and Table 2: Fund responses to capital flows

Read:

- **Table 1** shows the mutual fund sample from 1980 to 2006. The number of funds rises from 228 in 1980 to 1,789 in 2006.
- Mutual funds' ownership of the U.S. equity market rises from about 2.27 percent in 1980 to roughly 13.71 percent in 2006.
- **Table 2** estimates how mutual funds trade in response to flows.
- For outflows, the coefficient on flow is close to one, meaning funds sell close to dollar-for-dollar to meet redemptions.
- For inflows, the coefficient is about 0.62 in the simple specification, meaning funds invest only part of new money in old holdings.

Economic message: The data show why mutual funds are large enough to move prices and why investor flows mechanically translate into stock-level trades. Outflows generate forced sales; inflows generate partial but meaningful purchases.

Table 1
Summary statistics

Year	No. Funds	TNA (\$ Million)		Total Equity Holdings (\$ Million)		% Market Held	
		Median	Mean	Median	Mean	No. Stocks	% Held
1980	228	53.45	146.74	45.61	122.24	3,646	2.27
1981	226	53.66	137.71	42.11	109.31	3,543	2.21
1982	232	70.64	170.95	50.90	132.00	3,393	2.21
1983	255	97.41	222.14	79.74	182.20	4,173	2.74
1984	270	86.23	221.24	71.98	176.03	3,985	2.95
1985	297	114.12	275.98	89.48	222.04	3,845	3.08
1986	341	106.42	298.47	88.59	241.28	4,134	3.46
1987	376	87.00	286.30	74.03	238.41	4,544	3.89
1988	405	82.47	285.34	69.56	232.77	3,906	3.84
1989	440	95.08	340.49	77.91	265.36	3,798	3.92
1990	480	83.85	306.07	61.95	240.20	3,175	4.15
1991	579	100.23	379.32	79.85	309.56	3,548	4.78
1992	685	115.22	426.04	93.25	346.45	3,913	5.39
1993	925	105.56	442.40	90.00	350.65	4,663	6.54
1994	1,044	105.43	450.12	85.19	352.88	4,951	6.88
1995	1,168	134.35	610.98	112.60	488.36	5,338	9.02
1996	1,314	145.88	750.48	123.31	605.90	5,724	10.04
1997	1,480	163.42	933.60	135.21	774.02	5,858	11.07
1998	1,570	167.00	1,071.47	144.55	927.39	5,028	11.81
1999	1,686	187.52	1,307.48	164.05	1,139.49	4,958	12.95
2000	1,890	186.27	1,283.93	159.08	1,089.54	4,698	12.54
2001	1,915	155.22	1,018.79	133.73	882.57	3,670	13.36
2002	1,970	111.80	771.11	96.53	672.64	3,282	13.46
2003	2,001	146.05	976.25	128.51	852.98	3,760	13.54
2004	1,961	165.93	1,128.54	144.58	978.38	3,820	13.82
2005	1,918	196.90	1,251.72	169.84	1,067.81	3,884	14.02
2006	1,789	221.75	1,400.29	193.07	1,187.58	3,858	13.71

This table reports summary statistics of the mutual fund sample as of December in each year. The sample period is from 1980 to 2006. Because the focus of the paper is on the U.S. equity market, international, fixed income, and precious metal funds are excluded from the sample. Information regarding fund size, monthly returns, and capital flows is obtained from the CRSP survivorship-bias-free mutual fund database, and fund holdings data are obtained from the Thompson Financial's CDA/Spectrum database. The two data sets are then merged using the MFLinks file provided on WRDS. The table reports the following summary statistics: *No. funds* is the number of actively managed equity mutual funds at the end of each year; *TNA* is the total net assets under management reported by CRSP (in millions of dollars); *Total equity holdings* is the total dollar value of equity held by a mutual fund reported by CDA/Spectrum (in millions of dollars); and *% Market held* is the percentage of the U.S. equity market that is held by all mutual funds in the sample.

Table 1: Summary statistics

Table 2
Fund responses to capital flows

	The Outflow Sample				The Inflow Sample			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Intercept</i>	-0.059 (-6.62)	-0.029 (-1.32)	-0.022 (-0.85)	-0.022 (-0.88)	-0.032 (-3.42)	0.000 (0.02)	0.020 (1.22)	0.020 (1.21)
<i>flow_{i,t}</i>	0.970 (16.82)	1.028 (17.64)	1.107 (10.97)	1.107 (11.27)	0.618 (15.78)	0.737 (14.64)	0.858 (10.57)	0.855 (10.57)
<i>own_{i,j,t-1}</i>		0.429 (1.35)		-1.196 (-2.35)		-0.766 (-1.50)		-0.471 (-0.65)
<i>flow_{i,t} × own_{i,j,t-1}</i>				-20.588 (-0.58)		-12.431 (-3.74)		-1.669 (-0.51)
<i>liqcost_{j,t-1}</i>				-7.455 (-2.97)		-5.755 (-5.38)		-7.529 (-4.77)
<i>flow_{i,t} × liqcost_{j,t-1}</i>				-28.559 (-2.48)		-13.999 (-2.18)		-25.748 (-3.71)
<i>own_{i,t-1}</i>			2.171 (3.58)	3.924 (4.06)			-0.364 (-0.44)	0.212 (0.18)
<i>flow_{i,t} × own_{i,t-1}</i>			11.265 (1.32)	41.242 (3.10)			-21.337 (-3.20)	-19.235 (-2.58)
<i>liqcost_{i,t-1}</i>				-11.127 (-1.89)		-6.084 (-1.24)		-18.461 (-3.08)
<i>flow_{i,t} × liqcost_{i,t-1}</i>				-57.295 (-1.90)		-44.609 (-1.43)		-51.076 (-3.01)
Adjusted R ²	4.68%	6.31%	6.21%	6.43%	9.53%	10.07%	11.36%	11.46%
No. observations	1,207,060	1,044,623	1,207,060	1,044,623	2,462,355	2,215,898	2,462,355	2,215,898

This table reports regression analyses of mutual fund trading in response to capital flows. The dependent variable in all specifications is the percentage change in shares held by fund *i* in stock *j* from quarters *t-1* to *t* with split adjustments. The main independent variable of interest is *flow_{i,j,t}*, which is the net capital flow to fund *i* in quarter *t* divided by the fund's total net assets at the end of the previous quarter. Other control variables include *own_{i,j,t-1}*, the percentage of all shares outstanding of stock *j* that is held by fund *i* at the end of quarter *t-1*; and *liqcost_{j,t-1}*, the effective half bid-ask spread estimated from the Basic Market-Adjusted model as described in Hasbrouck (2006, 2009). *own_{i,t-1}* and *liqcost_{i,t-1}* are the portfolio-weighted average ownership share and

Table 2: Fund responses to capital flows

6.2 Table 3: The flow-induced price effect and Figure 1: Equal- and value-weighted stock return patterns of *FIT*

Read:

- **Table 3** sorts stocks by realized *FIT* and tracks returns around the formation quarter.
- Stocks heavily bought because of mutual fund inflows outperform stocks heavily sold because of outflows in the formation quarter.
- The formation-quarter effect reverses over the next two to three years.
- **Figure 1** visualizes the cumulative return of the high-minus-low *FIT* hedge portfolio.
- The value-weighted hedge portfolio rises in the formation period and then falls below zero by year three.

Economic message: Realized flow-induced trading creates temporary price pressure. Prices move in the direction of flow-induced demand and then revert, which supports the interpretation that the trades are not fully information-based.

Table 3
The flow-induced price effect

Panel A: The magnitude of FIT from quarters -4 to +4									
Decile	Qtr -4	Qtr -3	Qtr -2	Qtr -1	Qtr 0	Qtr 1	Qtr 2	Qtr 3	Qtr 4
1	1.27%	0.81%	0.48%	-0.24%	-5.50%	0.20%	0.59%	0.61%	1.21%
10	5.06%	5.80%	6.69%	6.62%	16.36%	4.06%	6.10%	5.25%	4.23%
10 - 1	3.79%	5.07%	6.21%	6.86%	22.27%	7.86%	5.81%	4.65%	3.02%
	(11.68)	(13.20)	(15.58)	(16.37)	(22.94)	(12.54)	(14.91)	(15.25)	(12.61)

Panel B: Equal-weighted returns to portfolios ranked by FIT									
Decile	Excess Return	3-Factor Alpha	4-Factor Alpha	Excess Return	3-Factor Alpha	4-Factor Alpha	Excess Return	3-Factor Alpha	4-Factor Alpha
Qtr 0 (Formation Qtr)									
1	0.39%	-0.33%	-0.54%	0.68%	-0.22%	0.06%	0.90%	-0.06%	0.32%
10	1.82%	1.06%	0.86%	0.66%	-0.02%	0.04%	0.49%	-0.31%	0.62%
10 - 1	1.73%	1.91%	1.59%	-0.01%	0.20%	-0.02%	-0.40%	-0.27%	-0.23%
	(7.77)	(5.31)	(7.38)	(-0.17)	(1.36)	(-0.10)	(-2.40)	(-1.40)	(-2.10)

Panel C: Value-weighted returns to portfolios ranked by FIT									
Decile	Excess Return	3-Factor Alpha	4-Factor Alpha	Excess Return	3-Factor Alpha	4-Factor Alpha	Excess Return	3-Factor Alpha	4-Factor Alpha
Qtr 0 (Formation Qtr)									
1	-0.22%	-1.05%	-0.82%	0.71%	-0.09%	-0.02%	0.81%	0.19%	0.32%
10	1.90%	1.26%	0.91%	0.66%	0.09%	-0.22%	0.21%	-0.31%	0.36%
10 - 1	2.12%	2.31%	1.76%	-0.08%	0.15%	-0.21%	-0.46%	-0.44%	-0.42%
	(5.96)	(6.78)	(5.11)	(-0.35)	(0.40)	(-0.93)	(-1.94)	(-2.85)	(-2.45)

This table reports calendar time returns to portfolios ranked by flow-induced trading (FIT). FIT is defined as the aggregate mutual fund flow-induced trading in a quarter divided by the total shares held by all mutual funds at the end of the previous quarter. The portfolios are rebalanced every quarter and are held for three years. Quarter 0 is the formation quarter. Panel A reports the magnitude of FIT in each of the nine quarters from one year before to one year after the ranking period. Panels B and C report the equal-weighted and value-weighted monthly portfolio returns in the following three years, respectively. To deal with overlapping portfolios in each holding month, I follow Jagadeesh and Titman (1993) to take the equal-weighted average return across portfolios formed in different quarters. Monthly returns with different risk adjustments are reported. The return in excess of the risk-free rate, the Fama-French three-factor alpha, and the Carhart four-factor alpha, t -statistics, shown in parentheses, are computed based on standard errors with Newey-West corrections of twelve lags. Estimates significant at the 5% level are indicated in bold.

Table 3: The flow-induced price effect

6.3 Table 4: Predicting future flows and Table 5: The expected flow-induced price effect

Read:

- **Table 4** shows that future mutual fund flows are forecastable from past four-factor alpha, market-adjusted returns, and lagged flows.
- Lagged alpha remains significant after controlling for lagged returns and flows.
- **Table 5** shows that expected flow-induced trading predicts returns.
- Stocks in the high $E[FIT]$ decile outperform low $E[FIT]$ stocks in the following year and underperform later.
- Funds with high $E[FIT^*]$ also outperform initially and then reverse.

Economic message: Predictability of fund flows turns realized price pressure into expected return predictability. The hump-shaped pattern is the central empirical signature of temporary flow pressure.

Table 4
Predicting future flows

	Fama-MacBeth			Pooled OLS		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Intercept</i>	0.028 (5.38)	0.028 (5.77)	0.010 (3.65)	0.016 (25.89)	0.014 (18.77)	0.007 (7.72)
<i>alpha_{i,t}</i>	4.827 (9.67)	1.766 (4.38)	0.953 (4.47)	4.232 (9.02)	2.453 (9.15)	1.330 (5.43)
<i>adjret_{i,t}</i>		0.396 (7.34)	0.229 (6.72)	0.202 (5.17)	0.089 (2.74)	0.089 (2.74)
<i>flow_{i,t}</i>			0.194 (8.78)		0.228 (17.21)	0.228 (17.21)
<i>flow_{i,t-1}</i>			0.102 (5.28)		0.109 (7.55)	0.109 (7.55)
<i>flow_{i,t-2}</i>			0.122 (6.29)		0.090 (6.03)	0.090 (6.03)
<i>flow_{i,t-3}</i>			0.033 (5.47)		0.029 (3.67)	0.029 (3.67)
Adjusted R^2	4.53%	7.70%	24.79%	5.25%	7.14%	19.83%
No. observations	98,264	98,264	95,285	98,264	98,264	95,285

This table reports forecasting regressions of future mutual fund flows. The dependent variable in all regression specifications is the capital flow to mutual fund i in quarter $t+1$ scaled by the fund's total net assets at the end of the previous quarter. The main independent variables include $\alpha_{i,t-1}$, the monthly Carhart four-factor alpha computed from the fund's returns in the previous year, and $adjret_{i,t-1}$, the cumulative market-adjusted fund return in the previous year. $flow_{i,t-1}$, $flow_{i,t-2}$, and $flow_{i,t-3}$ are lagged capital flows in the previous four quarters. The coefficients are estimated using both the Fama-MacBeth and pooled OLS approaches. Standard errors of the Fama-MacBeth estimates are computed with Newey-West corrections of four lags. For the pooled OLS approach,

Table 4: Predicting future flows

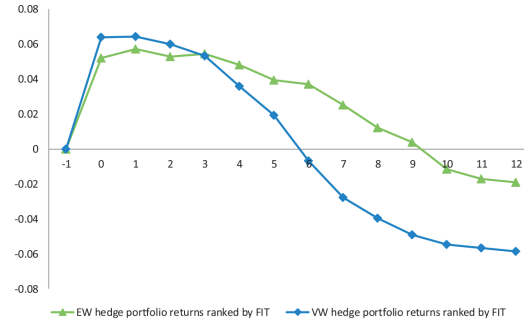


Figure 1

Equal- and value-weighted stock return patterns of FIT

This figure shows cumulative returns to the hedge portfolio ranked by FIT , which is defined as the aggregate mutual fund flow-induced trading in a quarter divided by the total shares held by all mutual funds at the end of the previous quarter. At the end of each quarter, all stocks are sorted into deciles based on FIT . These decile portfolios are then rebalanced every quarter and are held for three years. The hedge portfolio is a long-short portfolio that goes long in decile 10 and goes short in decile 1. Quarter 0 is the formation quarter. The curve with triangles shows cumulative returns to the equal-weighted hedge portfolio, and the curve with squares shows cumulative returns to the value-weighted hedge portfolio.

Figure 1: Equal- and value-weighted stock return patterns of FIT Table 5
The expected flow-induced price effect

Panel A: Stocks ranked by $E[FIT]$									
Decile	Excess Return	3-Factor Alpha	4-Factor Alpha	Excess Return	3-Factor Alpha	4-Factor Alpha	Excess Return	3-Factor Alpha	4-Factor Alpha
Qtr 1									
1	0.38%	-0.50%	-0.35%	0.53%	-0.40%	-0.11%	0.52%	-0.24%	0.84%
10	1.21%	0.43%	0.27%	0.97%	0.18%	0.24%	0.63%	-0.01%	0.54%
10 - 1	0.84% (3.96)	0.93% (4.13)	0.62% (3.51)	0.44% (2.63)	0.58% (3.30)	0.35% (2.26)	0.10%	0.23%	0.29% (0.81)

Panel B: Mutual funds ranked by $E[FIT^*]$									
Decile	Excess Return	3-Factor Alpha	4-Factor Alpha	Excess Return	3-Factor Alpha	4-Factor Alpha	Excess Return	3-Factor Alpha	4-Factor Alpha
Qtr 1									
1	0.58%	-0.17%	-0.03%	0.62%	-0.12%	-0.11%	0.71%	0.07%	0.87%
10	1.14%	0.54%	0.37%	1.02%	0.49%	0.26%	0.70%	0.11%	0.56%
10 - 1	0.56% (2.66)	0.71% (3.14)	0.41% (2.58)	0.40% (2.65)	0.58% (3.44)	0.37% (2.54)	-0.01%	0.04%	0.29% (0.22)

This table reports the association between expected capital flows to mutual funds and future stock and fund returns. Panel A reports calendar time returns to portfolios sorted by expected flow-induced trading ($E[FIT]$). Specifically, at the end of each quarter, all stocks are ranked into deciles based on $E[FIT]$, the aggregate expected mutual fund flow-induced trading divided by the total shares held by all mutual funds at the end of the quarter, where the expected capital flow to a mutual fund is estimated using the monthly four-factor fund alpha in the previous year. Panel B reports calendar time returns to various mutual fund portfolios. At the end of each quarter, all mutual funds are sorted into deciles based on $E[FIT^*]$, the portfolio-weighted average expected flow-induced trading. The portfolios in both Panels A and B are rebalanced every quarter and are held for three years. To deal with overlapping portfolios in each holding month, I follow Jagadeesh and Titman (1993) to take the equal-weighted average return across portfolios formed in different quarters. Monthly returns with different risk adjustments are reported. The return in excess of the risk-free rate, the Fama-French three-factor alpha, and the Carhart four-factor alpha, T -statistics, shown in parentheses, are computed based on standard errors with Newey-West corrections of twelve lags. Estimates significant at the 5% level are indicated in bold.

Table 5: The expected flow-induced price effect

6.4 Figure 2: Return patterns of $E[FIT]$ and $E[FIT^*]$ and Table 6: Mutual fund performance persistence

Read:

- **Figure 2** plots cumulative returns for hedge portfolios sorted by $E[FIT]$ and $E[FIT^*]$.
- Both curves rise during the first year and then decline, showing that the initial flow-driven return is temporary.
- **Table 6** conducts a horse race between fund alpha and $E[FIT^*]$.
- After controlling for $E[FIT^*]$, fund alpha has little remaining ability to forecast subsequent fund performance.
- Sorting first on alpha and then on $E[FIT^*]$ shows that $E[FIT^*]$ remains a strong predictor.

Economic message: The standard evidence for mutual fund performance persistence is largely absorbed by expected flow pressure. What looks like persistent skill may partly be the mechanical return effect of predictable investor flows.

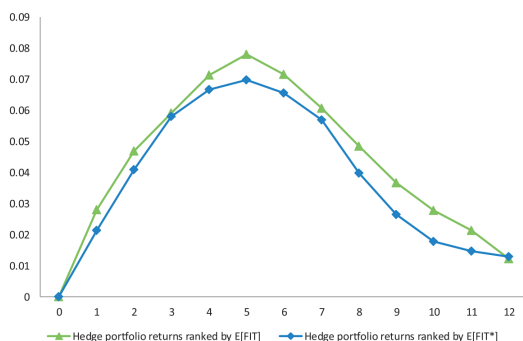


Figure 2
Return patterns of $E[FIT]$ and $E[FIT^*]$

This figure shows cumulative returns to the hedge portfolios ranked by $E[FIT]$ and $E[FIT^*]$. $E[FIT]$ is the aggregate expected mutual fund flow-induced trading divided by the total shares held by all mutual funds at the end of the quarter, where the expected capital flow to a mutual fund is estimated using the monthly four-factor fund alpha in the previous year; $E[FIT^*]$ is the portfolio-weighted average expected flow-induced trading. At the end of each quarter, all stocks are sorted into deciles based on $E[FIT]$. These decile portfolios are then rebalanced every quarter and are held for three years. The curve with triangles shows cumulative returns to the hedge portfolio that goes long in stocks in decile 10 and goes short in stocks in decile 1. Similarly, at the end of each quarter, all mutual funds are sorted into deciles based on $E[FIT^*]$. The curve with squares shows cumulative returns to the hedge portfolio that goes long in mutual funds in decile 10 and goes short in mutual funds in decile 1.

Figure 2: Return patterns of $E[FIT]$ and $E[FIT^*]$

Table 6
Mutual fund performance persistence

Panel A: Mutual funds first ranked by $E[FIT^*]$ then by alpha														
Quintiles of alpha	Quintiles of $E[FIT^*]$					Average	Quintiles of $E[FIT^*]$					Average		
	1	2	3	4	5		1	2	3	4	5			
Qtr. 1 (3-Factor Alpha)														
1	-0.17%	-0.06%	-0.06%	0.18%	0.36%	0.23%	0.05%	-0.09%	-0.07%	-0.08%	0.12%	0.27%	0.26%	0.05%
2	-1.23%	-0.90%	-1.03%	0.23%	0.33%	0.18%	0.08%	-0.64%	-1.18%	-1.27%	0.12%	0.28%	0.21%	0.02%
3	-0.29%	-0.02%	0.01%	0.09%	0.17%	0.27%	0.07%	-0.11%	-0.02%	0.09%	0.09%	0.21%	0.36%	0.03%
4	-1.25%	-0.49%	0.13%	0.50%	0.74%	0.27%	0.28%	-0.34%	-0.40%	-0.07%	0.05%	0.24%	0.23%	0.03%
5	-1.23%	-0.48%	-0.37%	0.60%	0.60%	0.28%	0.08%	-1.17%	-0.76%	-0.54%	0.15%	0.21%	0.19%	0.03%
6	-0.09%	-0.04%	-0.02%	0.05%	0.36%	0.47%	0.07%	-0.15%	-0.01%	-0.05%	0.06%	0.22%	0.34%	0.04%
7	-0.86%	-0.71%	-0.44%	0.27%	0.40%	0.24%	0.12%	-0.25%	-0.45%	-0.47%	0.18%	0.26%	0.13%	0.02%
8	-0.11%	0.01%	-0.01%	0.04%	0.37%	0.49%	0.07%	-0.01%	0.01%	0.06%	0.06%	0.49%	0.27%	0.03%
9	-1.02%	0.17%	-0.09%	0.09%	0.30%	0.24%	0.24%	-0.41%	0.36%	0.01%	0.02%	0.44%	0.28%	0.01%
10	0.06%	0.07%	0.05%	-0.14%	0.21%	0.07%	0.07%	0.04%	0.06%	-0.06%	0.39%	0.27%	0.07%	0.01%
11	0.36%	0.15%	0.71%	-1.71%	2.67%	0.18%	0.08%	0.35%	1.28%	1.07%	-1.06%	2.38%	1.21%	0.01%
Panel B: Mutual funds first ranked by alpha then by $E[FIT^*]$														
Quintiles of $E[FIT^*]$	Quintiles of alpha					Average	Quintiles of alpha					Average		
	1	2	3	4	5		1	2	3	4	5			
Qtr. 1 (3-Factor Alpha)														
1	-0.31%	-0.10%	-0.05%	0.01%	0.02%	0.22%	-0.15%	-0.04%	0.00%	0.05%	0.05%	0.18%	-0.03%	
2	-2.02%	-1.90%	-0.61%	0.07%	0.20%	0.21%	-1.10%	-0.94%	-0.53%	-0.40%	0.24%	0.22%	0.21%	
3	-0.17%	-0.05%	-0.05%	-0.01%	0.08%	0.21%	-0.04%	-0.09%	-0.05%	-0.04%	-0.01%	0.07%	0.19%	
4	-1.28%	-1.08%	-0.89%	-0.48%	0.24%	0.28%	-0.85%	-0.54%	-0.48%	-0.28%	0.07%	0.26%	-0.26%	
5	-0.17%	-0.05%	-0.05%	-0.01%	0.08%	0.21%	-0.04%	-0.09%	-0.05%	-0.04%	-0.01%	0.07%	0.19%	
6	-1.40%	-1.01%	-0.50%	-1.01%	0.27%	0.28%	-0.29%	-0.55%	-0.55%	-0.55%	0.17%	0.21%	0.06%	
7	-0.06%	-0.04%	0.01%	0.04%	0.47%	0.47%	-0.10%	-0.04%	0.02%	0.17%	0.30%	0.40%	0.06%	
8	-1.05%	-0.84%	0.50%	0.27%	0.66%	0.34%	-1.21%	-0.86%	0.36%	0.28%	0.76%	0.15%	0.40%	
9	0.17%	0.15%	0.34%	0.35%	0.01%	0.48%	0.05%	0.05%	0.27%	0.25%	0.40%	0.45%	0.23%	
10	0.15%	0.17%	0.55%	0.46%	0.73%	0.45%	0.73%	0.51%	1.21%	0.44%	0.24%	0.41%	0.26%	
11	0.45%	0.26%	0.45%	0.40%	0.40%	0.41%	0.24%	0.45%	0.27%	0.26%	0.47%	0.40%	0.40%	
12	0.45%	0.49%	0.24%	0.14%	0.39%	0.27%	0.43%	0.86%	1.19%	1.24%	0.24%	0.27%	0.26%	

This table reports the horse race between four-factor fund alpha and expected flow-induced trading to predict future fund performance. In Panel A, mutual funds are sorted into a five-by-five matrix first by $E[FIT^*]$ and then by the four-factor fund alpha. Panel B reports the same sequential sort in the reverse order: mutual funds are sorted into a five-by-five matrix first by the four-factor fund alpha and then by $E[FIT^*]$. $E[FIT^*]$ is the portfolio-weighted average expected flow-induced trading, where expected mutual fund flows are computed from the four-factor fund alpha in the previous year. The portfolios are rebalanced every quarter and are held for three years. To deal with overlapping portfolios in each holding period, I follow Jagadeesh and Titman (1993) to take the equal-weighted average return across portfolios formed in different quarters. Monthly returns with different risk adjustments are reported. The Fama-French three-factor alpha and the four-factor fund alpha, α_{FF} , estimates, shown in parentheses, are computed based on standard errors with Newey-West corrections of twelve lags. Estimates significant at the 5% level are indicated in bold.

Table 6: Mutual fund performance persistence

6.5 Table 7: The smart money effect and Table 8: Mutual fund performance regressions

Read:

- **Table 7** tests the smart money effect by sorting funds on lagged flows and $E[FIT^*]$.
- Lagged flows no longer predict future fund performance once $E[FIT^*]$ is controlled for.
- **Table 8** reports Fama-MacBeth fund-return regressions.
- $E[FIT^*]$ is significant in all specifications.
- Fund alpha and lagged flows predict future performance alone, but both are subsumed by $E[FIT^*]$ in the full specification.

Economic message: The smart money effect does not mainly show that investors identify skilled managers. It is more consistent with investors chasing performance, thereby creating predictable

flow pressure that temporarily boosts fund returns.

Table 7
The smart money effect

Quintiles of flow	Mutual funds independently sorted by $E[FIT^*]$ and flow									
	Quintiles of $E[FIT^*]$					Quintiles of $E[FIT^*]$				
	1	2	3	4	5	1	2	3	4	5
	Qtr. 1 (Excess Return)					Qtr. 1 (3-Factor Alpha)				
1	0.57%	0.71%	0.68%	0.79%	1.01%	0.45%	0.75%	-0.23%	-0.04%	-0.03%
2	(1.98)	(2.75)	(2.68)	(3.08)	(3.51)	(2.89)	(2.87)	(-2.13)	(-0.49)	(-0.54)
3	0.59%	0.69%	0.70%	0.74%	0.90%	0.30%	0.72%	-0.17%	-0.01%	0.00%
4	(2.10)	(2.71)	(2.70)	(2.91)	(3.02)	(1.91)	(2.70)	(-1.65)	(-0.10)	(-0.03)
5	0.65%	0.65%	0.62%	0.69%	0.93%	0.29%	0.71%	-0.10%	-0.06%	-0.06%
	(2.30)	(2.54)	(2.51)	(2.60)	(3.31)	(1.78)	(2.73)	(-0.80)	(-0.84)	(-1.18)
6	0.66%	0.66%	0.70%	0.71%	0.91%	0.25%	0.73%	-0.09%	-0.05%	0.00%
7	(2.33)	(2.54)	(2.73)	(2.69)	(3.07)	(1.60)	(2.77)	(-0.80)	(-0.66)	(0.01)
8	0.64%	0.71%	0.67%	0.80%	1.12%	0.49%	0.79%	-0.08%	-0.01%	-0.02%
9	(2.19)	(2.89)	(2.53)	(2.91)	(3.66)	(2.68)	(2.92)	(-0.65)	(-0.00)	(-0.39)
10	0.07%	-0.01%	-0.01%	0.00%	0.11%	0.03%	0.03%	0.14%	0.05%	0.11%
11	(0.83)	(-0.12)	(-0.19)	(0.00)	(1.25)	(0.61)	(1.56)	(0.44)	(0.18)	(0.40)
										(2.02)

This table reports the least square between lagged capital flow to mutual funds and expected flow-induced trading to predict future fund performance. At the end of each quarter, all mutual funds are sorted into a five-by-five matrix independently by $E[FIT^*]$ and lagged fund flows. $E[FIT^*]$ is the portfolio-weighted average expected flow-induced trading, where expected mutual fund flows are computed from the four-factor fund alpha in the previous year. The portfolios are rebalanced every quarter and held for one quarter. To deal with overlapping portfolios in each holding month, I follow Jegadeesh and Titman (1993) to take the equal weighted average return across portfolios formed in different quarters. Monthly returns with different risk adjustments are reported: the return in excess of the risk-free rate and the Fama-French three-factor alpha. t -statistics, shown in parentheses, are computed based on standard errors with Newey-West correction of twelve lags. Estimates significant at the 5% level are indicated in bold.

Table 7: The smart money effect

Table 8
Mutual fund performance regressions

	Fama-MacBeth Regressions of Quarterly Fund Returns						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Intercept</i>	0.050 (5.47)	0.053 (5.85)	0.053 (5.69)	0.054 (5.82)	0.049 (5.20)	0.051 (5.48)	0.051 (5.30)
$E[FIT^*]$	3.081 (3.06)				2.602 (2.35)	2.952 (2.93)	2.687 (2.43)
$\alpha_{hi,t}$		0.581 (3.82)		0.548 (3.64)	0.042 (0.24)		0.005 (0.03)
$\text{flow}_{1,t}$			0.012 (2.28)	0.010 (2.08)		0.004 (0.82)	0.004 (0.93)
$\text{expenses}_{i,t}$	-0.351 (-0.27)	-0.830 (-0.55)	-0.765 (-0.48)	-1.138 (-0.75)	-0.319 (-0.26)	-0.657 (-0.51)	-0.653 (-0.52)
$\log(\text{age}_{i,t})$	0.000 (0.17)	0.000 (0.47)	0.001 (0.63)	0.001 (0.90)	0.000 (0.37)	0.000 (0.65)	0.001 (0.84)
$\log(\text{numStocks}_{i,t})$	0.002 (3.58)	0.002 (3.95)	0.002 (3.72)	0.002 (3.78)	0.002 (3.27)	0.002 (3.44)	0.002 (3.02)
$\log(\text{TNA}_{i,t})$	-0.001 (-1.91)	-0.001 (-2.18)	-0.001 (-2.18)	-0.001 (-2.30)	-0.001 (-1.82)	-0.001 (-2.08)	-0.001 (-2.00)
$\text{turnover}_{i,t}$	0.002 (2.05)	0.002 (1.76)	0.002 (1.56)	0.002 (1.74)	0.001 (1.96)	0.002 (2.06)	0.001 (1.96)
Adjusted R^2	15.77%	11.03%	8.06%	11.91%	17.46%	16.53%	18.24%
No. observations	93,805	93,805	93,805	93,805	93,805	93,805	93,805

This table reports Fama-MacBeth forecasting regressions of future fund returns. The dependent variable in all specifications is the return of mutual fund i in quarter $t+1$. The main independent variables of interest include $E[FIT^*]$, the portfolio-weighted average expected flow-induced trading, where expected fund flows are computed from the four-factor fund alpha in the previous year; $\alpha_{hi,t}$, the Carhart four-factor fund alpha in the previous year; and $\text{flow}_{1,t}$, the capital flow to the mutual fund in the previous quarter. Other control variables include the expense ratio in the previous year, the fund age since inception, the number of stocks the fund is holding in its portfolio, the logarithm of the fund's total net assets of the fund at the end of the previous quarter, and the fund's turnover ratio in the previous quarter. Regression coefficients are estimated using the Fama-MacBeth approach. t -statistics, shown in parentheses, are computed based on standard errors with Newey-West corrections of four lags. Estimates significant at the 5% level are indicated in bold.

Table 8: Mutual fund performance regressions

6.6 Table 9: Stock price momentum and Table 10: Stock return comovement

Read:

- **Table 9** studies stock momentum using Fama-MacBeth regressions.
- $E[FIT]$ significantly predicts future stock returns.
- Controlling for $E[FIT]$ reduces the coefficient on lagged stock returns by roughly 25 to 50 percent, depending on the formation window and subsample.
- The flow mechanism is stronger in 1994-2006 and among large-cap stocks.
- **Table 10** shows that stocks sorted into the same $E[FIT]$ group experience correlated future flow-induced trading and excess return comovement.

Economic message: Flow-induced trading explains part of stock momentum and also creates common movement among stocks exposed to similar fund-flow demand. The mechanism matters for both expected returns and return covariances.

Table 9
Stock price momentum

Panel A: The full sample						
	$k=12$		$k=6$		$k=3$	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Intercept</i>	0.103 (2.81)	0.092 (2.36)	0.096 (2.63)	0.077 (2.01)	0.094 (2.58)	0.084 (2.34)
$E_t[FIT^k_j]$		0.085 (3.07)		0.145 (2.93)		0.250 (3.32)
$ret_{j,t-k:t-1}$	0.020 (4.06)	0.015 (3.31)	0.027 (3.59)	0.020 (2.82)	0.024 (2.29)	0.014 (1.40)
$ret_{j,t}$	-0.024 (-1.67)	-0.029 (-2.16)	-0.024 (-1.63)	-0.030 (-2.26)	-0.020 (-1.35)	-0.029 (-2.18)
$ret_{j,t-36:t-k-1}$	-0.005 (-3.19)	-0.004 (-3.05)	-0.004 (-2.64)	-0.004 (-2.56)	-0.004 (-2.54)	-0.004 (-2.54)
$bm_{j,t}$	0.005 (1.33)	0.005 (1.37)	0.005 (1.25)	0.005 (1.42)	0.006 (1.40)	0.006 (1.78)
$\log(mktcap_{j,t})$	-0.003 (-2.16)	-0.003 (-1.73)	-0.003 (-1.96)	-0.002 (-1.33)	-0.003 (-1.88)	-0.002 (-1.59)
$turnover_{j,t}$	-0.004 (-2.02)	-0.005 (-2.26)	-0.004 (-1.87)	-0.004 (-2.06)	-0.004 (-1.63)	-0.004 (-2.05)
Adjusted R^2	7.08%	7.85%	6.75%	7.85%	6.38%	7.88%
No. observations	198,692	198,692	198,692	198,692	198,692	198,692

Panel B: Subsample analyses ($k=6$)							
	1980–1993		1994–2006		Small-Cap Stocks		Large-Cap Stocks
	(1)	(2)	(3)	(4)	(5)	(6)	(7) (8)
<i>Intercept</i>	0.072 (1.37)	0.065 (1.29)	0.119 (2.54)	0.090 (1.65)	0.653 (5.53)	0.631 (5.16)	0.223 (5.84) 0.190 (4.28)
$E_t[FIT^k_j]$		0.106 (1.80)		0.203 (3.44)		0.158 (3.50)	0.175 (3.35)
$ret_{j,t-k:t-1}$	0.032 (2.77)	0.027 (2.75)	0.023 (2.44)	0.014 (1.92)	0.035 (4.82)	0.028 (4.62)	0.021 (3.10) 0.011 (1.57)
$ret_{j,t}$	-0.022 (-1.10)	-0.027 (-1.43)	-0.022 (-1.07)	-0.029 (-1.60)	-0.012 (-0.85)	-0.018 (-1.39)	-0.031 (-1.55) -0.041 (-2.36)
$ret_{j,t-36:t-k-1}$	-0.003 (-1.83)	-0.003 (-1.83)	-0.006 (-4.13)	-0.006 (-4.11)	-0.005 (-2.41)	-0.004 (-2.27)	-0.003 (-1.68) -0.003 (-1.62)
$bm_{j,t}$	0.004 (0.78)	0.003 (0.77)	0.007 (1.12)	0.008 (1.36)	0.006 (1.45)	0.006 (1.47)	0.002 (0.43) 0.003 (0.86)
$\log(mktcap_{j,t})$	-0.002 (-0.76)	-0.001 (-0.60)	-0.004 (-2.21)	-0.003 (-1.35)	-0.033 (-6.47)	-0.032 (-6.12)	-0.008 (-5.57) -0.007 (-3.93)
$turnover_{j,t}$	-0.007 (-2.23)	-0.007 (-2.31)	-0.001 (-0.29)	-0.001 (-0.41)	-0.006 (-2.49)	-0.006 (-2.73)	-0.001 (-0.44) -0.002 (-0.84)
Adjusted R^2	7.76%	8.44%	5.69%	6.99%	6.78%	7.55%	8.81% 9.96%
No. observations	65,047	65,047	133,645	133,645	89,255	89,255	109,437 109,437

This table reports Fama-MacBeth forecasting regressions of future stock returns. The dependent variable in all specifications is the quarterly stock return (i.e., the cumulative stock return in months $t+1$ to $t+3$). The main independent variables of interest include $E[FIT]$, the aggregate expected flow-induced trading divided by the total shares held by all mutual funds at the end of quarter t , where expected capital flows are computed from the market-adjusted fund return in the previous k months, and $ret_{j,t-k:t-1}$, the cumulative stock return in the previous k months. Other control variables include the lagged one-month stock return, lagged long-run returns, book-to-market ratio, logarithm of firm size, and average monthly turnover. Panel A reports regression coefficients for the full sample, which spans 1980–2006. Panel B reports the coefficient estimates for various subsamples. Small- and large-cap stocks are classified based on the median market capitalization of NYSE stocks at the end of the previous quarter. Regression coefficients are estimated using the Fama-MacBeth approach. t -statistics, shown in parentheses, are computed based on standard errors with Newey-West corrections of four lags. Estimates significant at the 5% level are indicated in bold.

Table 9: Stock price momentum

Table 10
Stock return comovement

Panel A: Comovement in monthly FIT					
Rank by $E[FIT]$	1	2	3	4	5
$FIT_{grp,t}$	0.626 (2.69)	0.413 (2.54)	0.169 (0.91)	0.573 (3.16)	0.751 (2.84)
$FIT_{ffind,t}$	1.072 (4.63)	0.993 (6.42)	0.978 (10.13)	0.907 (7.34)	0.962 (4.11)
Adjusted R^2	53.82%	52.69%	51.70%	53.17%	54.98%
No. observations	31,329	31,329	31,329	31,329	31,329

Panel B: Comovement in weekly stock returns					
Rank by $E[FIT]$	1	2	3	4	5
$ret_{grp,t}$	0.199 (6.84)	0.128 (6.00)	0.116 (7.60)	0.152 (8.30)	0.230 (8.78)
$ret_{ffind,t}$	0.419 (18.10)	0.487 (26.85)	0.507 (29.84)	0.483 (27.46)	0.402 (24.32)
$ret_{mkt,t}$	0.369 (13.50)	0.370 (17.02)	0.368 (17.13)	0.372 (14.66)	0.368 (9.45)
$ret_{smb,t}$	0.658 (25.14)	0.562 (35.46)	0.521 (33.36)	0.589 (31.65)	0.668 (17.26)
$ret_{hml,t}$	0.149 (7.07)	0.127 (6.75)	0.143 (6.72)	0.134 (5.02)	0.129 (4.28)
$ret_{umd,t}$	-0.100 (-8.78)	-0.068 (-6.72)	-0.055 (-4.62)	-0.045 (-3.24)	-0.027 (-2.36)
Adjusted R^2	27.22%	30.65%	31.73%	30.34%	26.74%
No. observations	39,170	39,170	39,170	39,170	39,170

This table shows correlation patterns in monthly flow-induced trading and weekly stock returns. At the end of each quarter, all stocks are sorted into quintiles based on $E[FIT]$, where expected fund flows are computed based on lagged four-factor fund alpha. I then conduct a time-series regression for each stock in each quarter using monthly fund flows or weekly stock returns in the following year. In Panel A, the dependent variable is the flow-induced trading in stock j in month t , and the independent variables are $FIT_{grp,t}$, the average FIT of the quintile portfolio to which stock j belongs, and $FIT_{ffind,t}$, the average FIT of the Fama and French (1997) 48-industry to which j belongs. In Panel B, the dependent variable is the return of stock j in week t , and the main independent variable is $ret_{grp,t}$, the value-weighted return of the quintile portfolio to which stock j belongs. Other control variables include the value-weighted return of the industry portfolio to which j belongs, as well as the market, size, value, and momentum factors. For $FIT_{grp,t}$, $FIT_{ffind,t}$, $ret_{grp,t}$, and $ret_{ffind,t}$, I exclude stock j from the calculation. I then take the cross-sectional averages of these regression coefficients for each quintile in each quarter. Finally, I report the time-series averages of these coefficient estimates and corresponding T -statistics, shown in parentheses, based on standard errors with Newey-West corrections of four lags. Estimates significant at the 5% level are indicated in bold.

Table 10: Stock return comovement

7 Short summary

- Mutual fund investors chase past performance, making flows predictable.
- Mutual funds respond to outflows by selling existing holdings almost dollar-for-dollar and respond to inflows by buying part of their existing holdings.
- Aggregating these trades across funds produces stock-level flow-induced trading, FIT .
- Realized FIT creates contemporaneous price pressure that later reverses.
- Expected FIT predicts stock returns: high expected buying pressure predicts positive short-run returns and negative long-run returns.
- Expected FIT^* predicts mutual fund returns in the same hump-shaped way.
- $E[FIT^*]$ subsumes past fund alpha and past fund flows in explaining future fund performance.
- The flow mechanism fully explains the paper’s measures of performance persistence and the smart money effect.
- Flow pressure also explains part of stock momentum, especially in the later sample and among large stocks.
- Flow-induced trading also contributes to excess comovement among stocks.

8 Conclusion

8.1 Core conclusion

The paper argues that predictable mutual fund flows create predictable price pressure. Investors chase fund performance; fund managers then buy or sell existing holdings; these trades move stock prices temporarily; and the resulting return continuation later reverses.

8.2 Economic intuition

The key friction is limited arbitrage. Even if flow-induced demand shocks are uninformed and partly predictable, arbitrageurs do not fully absorb them immediately. Prices therefore move in the direction of predictable demand and correct only gradually.

8.3 Incremental contribution revisited

The paper’s main contribution is to connect several return predictability facts to one institutional mechanism. Performance persistence, smart money, and part of momentum are not separate puzzles in this interpretation. They are different manifestations of predictable demand pressure created by retail flows into delegated portfolios.

8.4 Implications

- **For mutual fund evaluation:** performance persistence may overstate managerial skill if expected flow pressure is not controlled for.
- **For investor behavior:** chasing fund performance can create temporary gains but also subsequent reversals.
- **For asset pricing:** mutual fund flows are a source of nonfundamental demand that affects both returns and comovement.
- **For momentum:** part of momentum may come from institutional implementation of performance-chasing flows, rather than only from investor underreaction to information.

8.5 Limitations

The evidence is strong but still observational. The decomposition of trading into flow-induced and information-driven components relies on estimated partial scaling behavior. The mechanism is also fund-specific and may be stronger where mutual funds are dominant owners.

8.6 Takeaway

Return predictability can arise not only from information, risk, or investor psychology, but also from the plumbing of delegated asset management. Predictable flows create predictable trades; predictable trades create predictable, temporary prices.